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Logical analysis as a qualitative method I: Themes in old age and chronic illness

Abstract Qualitative methods have been criticized for their lack of explicit procedures and weakness in explanation. The present paper draws attention to a qualitative approach with a clear procedure and wide explanatory scope – logical analysis. Logic has been neglected as a way of organizing informants' accounts because it has been identified with a procedure for organizing theory. In this paper, the attempt is to reverse the emphasis, and to demonstrate a way of working with informants rather than involving in theoretical prescriptions. The approach is illustrated by some themes arising in old age and chronic illness.

First, two conceptions of illness which have been distinguished in the work of Claudine Herzlich are analysed as systems of premises and consequences. Next, the procedures involved in logical analysis, and the explanations and practical uses which arise from it, are illustrated in two case studies. These cases are part of a larger body of data collected, amongst other things, for the purpose of eliciting the logic of old people's thinking.

The emphasis is on illustrating the method; and general issues which arise from it, including the analysis of people who entertain contradictory systems of thought, are postponed to a second paper.

Introduction

Qualitative methods have held an enduring place in sociology despite considerable and often well-founded criticism. Some of that criticism has been directed to identifying the appropriate topics for such methods, and to restricting their invasion of inappropriate areas. Other comments have pointed out the vagueness of qualitative studies about their procedures of data collection and analysis, and their weakness in providing general explanations; and a growing literature has discussed ways of repairing these faults.¹ A method which seems to avoid many notorious faults, and to illuminate a specific topic, is offered in this paper.

Among the appropriate topics for qualitative methods it is normal to include questions about reasons as opposed to situational causes,

about values as opposed to institutions, or about cultural meanings as opposed to structural constraints. The relation between internal meanings and external situations, for instance, has become the focus of the popular 'career' study which traces the experience of individuals over time as they come into contact with a major public institution. Less popular than the subject of 'careers' though, and in many ways unrecognized because it may be seen merely as a marginal aspect of a 'career' study, is the equally important topic of how internal meanings are related to one another.

This topic, the interconnection of meanings, is that for which a qualitative method is suggested here. The method proposed is an adaptation of something long established in another discipline — formal logical analysis. Logic has the great advantage in this respect that its procedures of analysis are clear, and its canons of explanation are perhaps the oldest in the business. Certain humble uses of these virtues, though, have been obscured by the sophisticated functions of symbolic logic in philosophy and the natural sciences. In philosophy, symbolic logic has been used to exhibit the nature of logical form when denuded of content. In the natural sciences, symbolic languages have been used to build theoretical assumptions and to derive new consequences from them. Resistance to these models of analytic procedure has consumed much ink in sociology, with good reason; and these are not the uses proposed here. The proposition here is for logic to be used without its symbolic paraphernalia simply to reveal the logical shape of an informant's ideas. Instead of measuring the informant against the researcher's logic, the research attempts to elicit the informant's logic. But the procedure is more easily demonstrated than discussed, and the emphasis of this paper is solely on demonstration.

The method will be worked through on the topic of old age and chronic illness, leaving to a later paper the theoretical issues which arise. Ageing and disablement have been studied extensively in recent years, but the literature mainly deals with the external circumstances of the old people concerned. 'Career' studies, in which the internal experience of those circumstances might have been illuminated, have on the whole been lacking; and while work has now begun to emerge on the world of older people as they currently perceive it,² some sociologists have rightly wished to see in addition a proper value given to the influence of biography on that world.³ To some, though, biographical study on the themes of old age, for all its evident attraction, has seemed a hopeless ideal, and it would indeed be hopeless if one sought to establish causal relations between present circumstances and events as they were long ago, rather than as an old person now thinks that they were. But what people think of past events — the set of conclusions which they draw from remembered experiences — is itself of great importance, and

biography in this sense, as a set of premises inferred from the past, is one influence on conceptions of illness for which logical analysis is likely to be appropriate.

Definitions of illness and disability also vary, though, under the influence of a whole constellation of current cultural conceptions; and an excellent example of such cultural variations is the study by Claudine Herzlich on middle-class Parisians.⁴ I have capitalized on her work by subjecting to logical analysis some of the conceptions of illness which she has distinguished, and this will give a preliminary example of what is entailed in the approach. It is not merely as an illustration, though, that her work is important; I hope also to show that it gives rise to a general explanatory scheme of considerable power.

In order to reveal the logical form of Herzlich's material, I shall render the attitudes described by her as a sequence of propositions. In doing so, it is impossible to 'shrink' the number of words used by the author without losing meaning, but this is permissible, as a generalization is permissible, in order to handle wider connections. The point is therefore not faithfulness to the text at all costs, but the extraction of essential sequences of thought which may subsequently be compared with other material. However, as realism is important in proceeding from connection to connection, I will disregard some conventions, particularly in the text, in the way I state propositions and logical sequences. In the figures, though, and in the conclusions on each case study, I will make a more formal presentation of premises, although only to a degree sufficient for sociological purposes.⁵

An illustration of logical analysis from the literature – Herzlich's study

The study by Claudine Herzlich on health and illness is in fact quite an old one, although its publication in English has been more recent, and its lack of influence in British and American medical sociology to date is undoubtedly due in part to its unfamiliar approach. It is also due, though, to the rather high level of generality at which its conclusions appear to operate, and the lack of any context in the way of illness behaviour or family negotiation which might lend realism to what are essentially the contents of a speculative conversation between the interviewer and the respondent. However, credibility is lent to her conclusions by their correspondence with, and addition of detail to, themes which occur in more orthodox medical sociology. The concept of 'normalizing' for instance, has close resemblances to Herzlich's concept of illness as an occupation', and the concept of 'disassociation' appears to be mirrored in her 'illness as a destroyer'.⁶ Herzlich also,

notably, develops a third category, 'illness as a liberator', which likewise merits attention, but in this paper I will focus only on the two more orthodox categories.

Herzlich's category of 'illness as a destroyer' then, begins with premises something like these:

- If I am active, then I am not ill.
- If I am myself, then I am active.
- I have something to offer if, and only if, I am active.

The consequences of these three premises, which are so innocuous at first sight, are surprisingly extensive, and soon begin to imply definitions of the sick, behaviour towards the sick, and behaviour as a sick person. Notably, though, at this stage only one of the premises actually mentions illness. Here are some initial consequences of recognizing 'I am ill':

- I am not myself (or as it is sometimes expressed, 'I' am 'dead').
- I do not have something to offer.

then

- If one does not have something to offer, then one cannot repay help.

therefore

- I cannot repay help.

but usually

- You are trying to help me.
- Either one repays help, or else one is a child.

so

- Go away and leave me alone.

or

- I am a child.

Parallel consequences will of course follow from the recognition 'You are ill', and result in avoidance of the sick, or a patronizing or pitying treatment of their needs. The terrible alternatives which this line of connection throws up may then lead anybody who holds these premises to answer the question 'Am I ill?' or 'Are you ill?' with a re-affirmation of the first premise as a *solution* to the question, one which, however, in this context and use, acquires a new connotation:

- If I am active, then I am *not* ill.

Then other commonplace premises are at once brought into play:

I need a medical diagnosis if, and only if, I am ill.

I act ill if, and only if, I am ill.

From which it follows that

I do not need a medical diagnosis.

I am not seen taking medicine.

But once it is no longer possible to deny 'I am ill' in the face of the definitions of all about one, some other premises may be brought into play with further interesting consequences.

If 'I' am 'dead', then I have nothing to fear from death.

If I have nothing to fear from death, then the doctor can do what he likes.

These two polar reactions, involving refusal of medical information, surreptitious preventive efforts, irritation with help and fierce denials on the one hand, and helpless compliance, childishness, and carelessness of the outcome on the other, present I suspect, an all-too-recognizable portrait of one kind of patient – a 'bad' patient when there is disease and the doctor is diagnosing it, or when the disease is chronic, but a 'good' patient when there is no serious disease, since he will not complain of ambiguous symptoms, and in some respects a 'good' patient on whom to operate. Thus 'bad' or 'good' is merely contingent, and this patient's behaviour emerges throughout as the remorselessly rational consequence of sensible propositions. The whole system of thought is summarized formally in Figure 1.

'Illness as an occupation' is in some respects the antithesis of 'illness as a destroyer', but as a system of thought it derives from premises which are not merely antithetical. Indeed a number of important premises which have nothing specific to do with illness, can be shared by both systems. These are:

If I am myself, then I am active.

I have something to offer if, and only if, I am active.

If one does not have something to offer, then one cannot repay help.

Either one repays help, or else one is a child.

In some ways the other premises of this way of thinking may also look suspiciously like those which made the definition of illness as 'a destroyer'. Two of them might run something like this:

If I am healthy, then I am strong.

If I am strong, then I am active.

But – and this is the key to this mode of seeing the world – it does not follow from *these* premises that

If I am active, then I am not ill.

Figure 1 Logical representation of 'illness as a destroyer'

Each consequence depends on the premises (or on consequences which in turn have become premises) which are numbered in brackets on its right-hand side.

'Illness as a destroyer'

General premises

- 1 If I am active then I am not ill.
- 2 If I am myself then I am active.
- 3 I have something to offer if, and only if, I am active.
- 4 If I do not have something to offer, then I cannot repay help.
- 5 If I accept help, either I repay help, or else I am a child.
- 6 I need a medical diagnosis if, and only if, I am ill.
- 7 I act ill if, and only if, I am ill.
- 8 If 'I' am 'dead' then I have nothing to fear from death.
- 9 If I have nothing to fear from death then the doctor can do as he likes.

Definition of the situation

10 I am active

10 I am ill and I must accept or refuse help.

Consequences

- | | | | |
|--------------------------------------|--------------|--------------------------------------|---------------|
| 11 I do not need a medical diagnosis | (10),(1),(6) | 11 I am not myself 'I' am 'dead' | (10),(1),(2) |
| 12 I do not act ill. | (10),(1),(7) | 12 I do not have something to offer | (10),(1),(3) |
| | | 13 I cannot repay help | (10),(12),(4) |
| | | 14 I refuse help or I am a child | (10),(13),(5) |
| | | 15 I have nothing to fear from death | (10),(11),(8) |
| | | 16 The doctor can do what he likes | (10),(15),(9) |

Rather, illness first makes its appearance in a basic tenet of the following kind:

If I am ill, then I fight.

And because, if I fight, I am active, it is plain that health here can

coincide with illness and include it, making, as Herzlich's respondents described it, a *reserve* of health from which they gather strength to fight the illness.

The consequences of this one vital shift of premise are immediate, for because of this shift it remains possible to say that:

I have something to offer.

I am myself.

I can repay help.

I am not a child.

and relations are free to continue to develop between myself as a patient and my healthy acquaintances and between myself as a healthy person and somebody who is sick.

The new premise also influences the definition of doctors, and the value of the sick person's own activities. The central idea of 'fighting' has a number of corollaries:

If I fight, then I fight well.

If I fight well, then I cut losses and develop any resources I can find.

Now although there is nothing in the present way of thinking to prevent people from asking questions about their illness and taking medicines openly, there is also something which naturally impels them in this direction. It can be stated like this:

If I do not demand explanations, I do not find out what I am fighting.

If I do not find out what I'm fighting, I do not fight well.

Hence it is characteristic of this system of thought that the doctor is cross-questioned and his orders constantly appraised for clarity and consistency. Naturally, a patient who thinks like this can be very difficult, although he is a model patient from the point of view of having 'spirit'. He also not only saves what he can from the wreck, but searches for new resources, whether activities or social connections, which can help to sustain him. In this sense,

If I fight well, then I am strong.

And strength is seen to come not only from health, but from the process of the struggle itself. Again, the whole conception of health and illness in this way of thinking can be represented formally, as in Figure 2.

Two partly contrasting conceptions of illness have now been analysed. But the analysis, at this stage, may still bear a resemblance to the logical construction of theory in the abstract, because no reference has been made to the thinking of specific informants. Nevertheless the propositions differ from theoretical propositions in having, for the most part, a first person subject, for all that the subject is not identified.

Figure 2 Logical representation of 'illness as an occupation'

Each consequence depends on the premises (or on consequences which in turn have become premises) which are numbered in brackets on its right-hand side.

'Illness as an occupation'

General premises

- 1 If I am myself I am active.
- 2 I have something to offer if, and only if, I am active.
- 3 If I do not have something to offer, I cannot repay help.
- 4 If I accept help, I repay help or else I am a child.
- 5 If I am healthy then I am strong.
- 6 If I am strong then I am active.
- 7 If I am ill, I fight and fight well.
- 8 If I fight well, I cut losses and develop any resources I can find.
- 9 If I fight, I am active.
- 10 If I do not demand explanations, I do not find out what I'm fighting.
- 11 If I do not find out what I'm fighting I do not fight well.

Definition of the situation

- 12 I am ill and I must accept or refuse help.

Assumptions consistent with the premises

- 13 I am myself.
- 14 I can repay help

Consequences

- | | |
|---------------------------------------|-----------------------|
| 15 I have something to offer | (12), (7), (9), (2) |
| 16 I can avoid being a child | (12), (14), (4) |
| 17 I cut losses and develop resources | (12), (7), (8) |
| 18 I demand explanations | (12), (7), (11), (10) |

The difference is important, for it symbolizes that the truth values of each proposition will not ultimately be determined by the researcher's language but by the informant's. To compare the way Herzlich's informants actually thought, with the logic which has been presented to depict their thinking, all I can now do is to refer readers to her text. At this point in the present paper, though, the most fruitful line will be to present some of my own data, and show the operations of logical analysis on tape transcripts. A procedural demonstration will be one aim of these case studies; but I will also stick closely to the topic of Herzlich's analysis – the two notions of illness – in the hope of showing that these patterns of thought can, with the aid of logical analysis, offer general explanations of a kind quite other than those for which logic has been used in the past.

Two case studies in systems of thought about illness

The Scottish data which I will now draw on have been collected, amongst other considerations, for the purpose of eliciting the logic of informants' ideas. As regards the choice of informants, people over sixty years of age were approached in a working-class and a middle-class area through the medium of social clubs. From these initial contacts, introductions were obtained, carefully varied by age, sex, and marital status, and most particularly by whether they belonged to clubs or not, with the aim of maintaining proportions in each category similar to those found in national data. The interviewing of these people has all been done by the author, and there have been a number of repeated interviews, and some informal social contact as well. Everybody was asked to talk informally about how health had affected, and was affecting, their lives. The early questioning elicited health history and residential history, but thereafter it proceeded on positions already taken, seeking to clarify them particularly in relation to future expectations of illness and dying, and planning decisions related thereto. Afterwards, fresh thinking and behaviour related to health and illness were picked up in later interviews. Also, as a further check on the interpretation, the design has facilitated cross-references by some informants to the behaviour of others. The study is still developing, but at the time of writing the pool of data included tape transcripts or field notes on conversations with 41 working-class people in 30 households, and 28 middle-class people in 22 households.

From this material I have selected two case studies to develop the content of Herzlich's insight in relation to logical analysis. The first is a widow in her late seventies from the working-class area whose thought approximates closely to the type christened by Herzlich, 'Illness as a destroyer'. The second is a widow in her late sixties from the middle-class area whose way of thinking approximates to the other type, 'Illness as an occupation'. Both cases are chosen because they present these systems of thought in a form which is virtually complete and unqualified. Both are people who strike one as very much 'all of a piece'. They may puzzle over decisions and conclude afterwards that they made mistakes, but they continue to identify both their correct decisions and their mistakes in the same coherent terms. In this, of course, they differ from the sort of person who, like most of us, is going to present the obvious problem, as well as the most interesting opportunity, for the analysis of premises: the person who entertains contradictions. This case will, however, be dealt with in another paper. The present two case studies will develop the portrait of the two systems of thought about illness, and will reveal most of the initial

methodological issues in analysing the logical implications of illness in the mental world of older people.

My treatment of the two cases varies progressively. In the first study I pay special attention to the process of learning the logic of the informant's idioms and translating them into the terms of each system of thought. Herzlich's informants were middle class, because she felt that this class had an active role in 'the genesis, transformation and renewal of ideologies'.⁷ Since this suggestion implies that her analysis might work best for the highly articulate, I have chosen for this study what ought to be the difficult case – the working-class widow, Mrs Stone. Also, in the conclusions on Mrs Stone, I spend time on discussing the formal requirements of this type of explanation. The formal requirements of the explanation and the process of translation are then largely taken for granted in the second study of Mrs Hannay, making her study the shorter, although the volume of material collected on Mrs Hannay was actually the more extensive. In the conclusions on each study, I also draw out possible practical uses of the method.

(a) *Mrs Stone*

Mrs Stone is the working-class widow whose philosophy was close to that of 'illness as a destroyer'. Very early in my acquaintance with her, when she was reviewing the places she had lived in her lifetime, her mind moved to the association of her present house with her husband's slow death from cancer. Apart from recounting the stages of his decline and the process of discovering its cause, she also enlarged on a detail which later proved significant. For the sake of easy assimilation I emphasize terms common to several later contexts, and these will guide the analysis.

'Although he came to this house he didn't like this house because there was no view. He used always just to sit in this chair and there's nothing for them to look at. When we were in Charlotte Street you used to go through to the room, you'd see the old tap and people passing and things like that. He just had nothing here for him.'

This was her first mention of illness, but soon she referred also to a sister who had 'been ill ever since her husband died, she's a poor thing really'. The expression 'a poor thing' is a common one in Scotland, and the connotations which follow are typical:

Q: 'Has your sister taken it well – her husband dying?

A: 'Oh no she's gone the other way, she's pined terrible and she's not well, not a well person at all. Now Grace my other sister she's like myself, she's go-ahead, she's go-ahead.'

Q: 'Gets on with life?'

- A: 'She gets on with life, but not Nellie. She's scared, and she's scared still to *go out* at night' . . . (and later) 'But she will not *go out* — we ask her along with us to Grace and I, but she gets tired, very, very tired and there in her face it'll be all in. Still that yet.'

And later:

- Q: 'But there are good and bad ways . . . ?'
 A: 'Good and bad ways — and she's took the bad way.'
 Q: 'What's the good way?'
 A: 'Well just *going on in your own way*, *going* to things like I'm doing, instead of *sitting* and *moaning*.'

By this time the conversation was picking up a theme which made explicit the generality of her terms:

sitting
 going out

Again further terms to be taken up in other contexts are italicized:

- Q: 'Are there things you do to keep yourself healthy?'
 A: 'No, just, I'm never off my feet. I'm always going about *doing work* and *helping* somebody else. As a rule *I'll* go and see *them*, and I still do it yet. I've always been on the go, and that's the thing to keep you going. If you sit down to it, well you've had it really, you won't *get up*. At my age or even at seventy, some of them say they're old at sixty, oh I canna *get up*, I have to sit in a chair, and this, and this; and I tells them, I says, you should move about, I says, and get out. Well I go to the dinners, *helps* . . . well, I say the old people but (laughs) . . .

Now can one put all that Mrs Stone has said so far in a few sentences? I suggest that some of her basic propositions are:

I moan only if I am sitting
 I sit only if I don't get up.
 I don't get up only if I don't go out.

Or to put it still more shortly:

I moan only if I don't go out.

This proposition has been expressed in several contexts. In the mean time though, some further propositions may be added which have not been so obviously rooted in a variety of contexts. To the interviewer these sentences become part of a hypothesis, in which he asks himself: do these propositions cross contexts?

If I am helping somebody, I am doing work.
 If I am helping somebody, I am going out.

If I am doing work, I am healthy.

If I am a well person I go on in my own way.

If I am going on in my own way I go out.

And for the sake of a translation into the language used by Herzlich's informants the suggestion may be added that:

If I am myself, I am going on in my own way.

It may be useful at this point to consider the parallels already apparent between Mrs Stone's terms and that of 'illness as a destroyer':

'Illness as a destroyer'

Mrs Stone

being myself

going on in my own way, being a well person

being active

going out

having something to offer

helping somebody

The question is now what Mrs Stone understands by being healthy. Does she regard it as the opposite of illness? Does she, as one might suspect from the importance in her thought of 'going out', subscribe to the crucial premise:

If I am active, then I am not ill.

or in her own terms

If I am going out, then I am not ill.

If she does, the following consequences immediately follow from the suggestion 'I am ill':

I don't go out.

I don't go on in my own way.

I am not a well person.

I am not doing work.

I am not helping somebody.

I moan.

And in the metaphor used by Herzlich's informants:

'I am 'dead'.

Here now is Mrs Stone replying to a suggestion of this sort. I emphasize the crucial terms.

Q: 'Have you ever thought how you'd cope with being ill?'

A: 'Oh I don't know if I'd be *ill* or *die* (laughs). Oh I wouldna like to be ill.

I pray that I'm able to do my *work*, and that myself, without troubling my family. I pray for that, I've always done that.'

Q: 'So really you would like to go just bonk?'

A: 'Yes — just bonk' (laugh).

And later on:

Q: 'You'd just like to be overtaken by it (*death*)?'

A: 'Oh, just what it is. Just say oh don't say I'm about to be *ill* . . . I wouldna like the *lying*. Oh don't give me the lying . . . oh I never have my bed, my bed — there's an awful lot goes to bed for the least thing, I know that. Not me. If I . . . I had a *sore back* two or three weeks back — *it didn't stop me from going out* or anything like that . . . I've never stayed in. It's a *punishment* for me to stay in.'

There is little doubt from this that:

If I am going out, then I am not ill.

It is only necessary now to ask whether Mrs Stone subscribes to the rule of reciprocity, with the consequences in her system of rejection of help or else a sense of being childish; and whether she denies serious illness when it is present, and subscribes to the equation of medical diagnosis and overt medical regimes with illness, with the consequences in her system of avoiding medical diagnosis and using only unobtrusive remedies.

There was some indication even in this first conversation that in both cases she regarded these consequences as valid. Indeed, as regards the rejection of help, analysis of the whole system of her thought cleared up a passage which at the time, and in my interviewer's language, left me with an apparently insoluble contradiction. The passage concerned help after bereavement:

Mrs Stone: 'It's really terrible. There's any amount of people coming to the door and sympathizing with you and one thing and another.'

Q: 'Is it a bit too much having . . . ?'

A: 'Too much, yes, it is too much.'

Q: 'Did you find that?'

A: 'Yes I did.'

Q: 'You wanted to be on your own?'

A: 'I wanted to be on my own.'

Q: 'That's interesting because people sometimes say that we leave people alone too much now?'

A: 'Aye, oh I think so.'

Q: 'You think that's true too?'

A: 'Yes it is really.'

As will soon be evident, my floundering attempt at repairing the contradiction I had created was inept:

Q: 'You want a happy medium.'

A: 'Yes, aha.'

In fact Mrs Stone did mean both propositions of the apparent contradiction. She meant:

- 1 It is too much so many people *coming* and *helping*.
- 2 One should not be *sitting* on one's own.

The implied third possibility became clear when she discussed her sister's bereavement later. The sister too didn't like helpful visitors:

Q: 'Did she get too many sympathizers, or too few?'

A: 'Yes – she didn't like that either.'

Q: 'What?'

A: 'Sympathizing.'

At the same time, so we have seen, her sister was 'a *poor thing* really' and sat at home on her own. But this double bind only took effect because 'she will not *go out*' – the crucial third possibility. Hence it is clear that Mrs Stone does reject help, and we may draw the following further parallels with 'illness as a destroyer':

<i>'Illness as a destroyer'</i>	<i>Mrs Stone</i>
wishing to be left alone	wishing to be on my own
being helped	being sympathized with
being a child	being a 'poor thing'

Finally it was apparent even in our first conversation that Mrs Stone was inclined to deny the possibility of serious illness:

Mrs Stone: 'I've never had it (symptom) two years now, and I had it badly afore, and I don't know what's happened to it but – still alive and going strong.'

Q: 'Do you think that as you get older your health is bound to get worse?'

A: 'I haven't. I have not.'

Q: 'So you think that as you get older still it won't make any difference?'

A: 'No . . . no. No if I keep the way I'm doing, and my birthday's in February, I'll be seventy-seven then (laughs) . . . take it as it sounds, aye . . . seventy-seven then; ach well.'

Not surprisingly, as far as medicaments went Mrs Stone favoured taking tablets and keeping going.

Events were, however, to reaffirm the tendency to denial which her premises appeared, in this first conversation, to imply. She found a lump. For three months, during which time she went on holiday with her sister and also saw several sisters and a daughter regularly, she said nothing about it to anyone. To members of her circle, nevertheless,

it was apparent that she was worried. Finally it became very painful, and at last she asked a neighbour, on a casual meeting in a lobby, what she thought it might be. The neighbour suggested a simple curable condition. 'So that made me feel better.' She then told her daughter, with the predictable result that her daughter had her 'straight up to the doctor' and the doctor had her straight into hospital where the specialist 'called me everything', saying that she had nearly made a simple condition dangerously complicated by her delay. Cured, and back at home, she continued nevertheless to maintain silence in front of her neighbour on what she thought it might have been. Only later, and *tête à tête* with the neutral interviewer, did she admit what had been obvious to everyone: 'I thought it was cancer.'

Conclusions on Mrs Stone

I have sought to show how one proceeds with a logical analysis of Mrs Stone's thoughts and behaviour in her own terms. After analysis, the next step is explanation. In a formal explanation of this kind, taking any one consequence in Mrs Stone's behaviour or thought, an economical set of her premises can be stated which explain the consequence, given three main conditions:

- (a) that the investigator has learnt Mrs Stone's language and logic correctly;
- (b) that there is no reason to invalidate any of Mrs Stone's premises as rationalizations, on the grounds that they cost her nothing and do no work other than justifying one specific response;
- (c) that no premise antecedent to the premises stated countermands their application to emerging consequences.

Subject to these three conditions, Mrs Stone's explanation of her statements and behaviour has a *prima facie* validity, though it makes no claims to be complete. Its connections with other premises, and similarly with situations which give rise to the premises held, can be elaborated indefinitely. But thus far it has a fair claim to be valid.

Next, in order to see Mrs Stone's explanation as an instance of a general system of thought, it is necessary to translate, making equivalences between the propositions and terms of her system and the general system. This process takes place subject to a fourth condition:

- (d) the investigator's translations are acceptable.

All these conditions for the validity of the explanation can now be reviewed. It is convenient to take condition (a), the investigator's learning of the informant's logic, and condition (d), his translations into the general system, together, since both are essentially tested by inspection. My summary of Mrs Stone's terms and logic, and my translations into the general system of 'illness as a destroyer', can be given as follows:

'Illness as a destroyer'

- 1 If I am active then I am not ill.
- 2 If I am myself then I am active.
- 3 I have something to offer if, and only if, I am active
- 4 If one does not have something to offer, then one cannot repay help.
- 5 Either one repays help, or else one is a child.
- 6 I need a medical diagnosis if, and only if, I am ill.
- 7 I act ill if, and only if, I am ill.
- 8 If 'I' am 'dead' then I have nothing to fear from death.
- 9 If I have nothing to fear from death then the doctor can do as he likes.

Mrs Stone

- 1 If I am going out then I am not ill.
- 2 If I am going on in my own way then I am going out.
- 3 If I am helping somebody I am going out.
- 4 (Assumed and proven by consequences.)
- 5 If one is helped and cannot help somebody, one is a 'poor thing'.
- 6 (Assumed and proven by consequences.)
- 7 (Assumed and proven by consequences.)
- 8 I don't know if I would rather be ill or die.
- 9 (Not canvassed.)

Having presented these implications and equivalences for inspection, I turn to the remaining two conditions of validity. Condition (b) forbade rationalizations. The crucial differentiating premise (1) is not a rationalization, because it follows from a set of premises around which Mrs Stone's current life demonstrably revolves:

- If I am helping somebody, I am doing work.
- If I am helping somebody, I am going out.
- If I am doing work, I am healthy.
- If I am healthy, I am not ill.

Hence condition (b) above is fulfilled. Condition (c) prohibiting a hidden qualifying premise, is assumed unless falsified by actual consequences. But the consequences of these premises, both present and future, for Mrs Stone, in terms of rejection of help, moral diminution of the sick receiving help, denial, avoidance of medical diagnosis, and preference for unobtrusive medication, have been demonstrated.

The formal properties of the explanation have thus been exposed, and one may now begin to talk about its practical applications. The case of Mrs Stone illustrates some practical points. The doctor who 'called her everything' for delaying so long was no doubt hoping to influence her behaviour in the future. In this hope he was almost certainly mistaken. Mrs Stone was indeed mildly abashed by the drubbing she received both from the doctor and her daughter. But although she was not indifferent to approval, she could not pinpoint how her

premises differed from theirs and was if anything made more inclined to do what she would logically do anyway – keep silence. Her only comment was:

‘Funny isn’t it. You don’t want to let things out.’

What the doctor had to speak to, if he wished to gain Mrs Stone’s ear, was one of her primary and distinctive premises. He would ignore the intervening premises, such as the role of reciprocity, which she shares with other systems of thought, and he would look for the origin of the crucial premise:

‘If I am going out, then I am not ill.’

The premise on which he could probably most usefully focus is:

‘If I am doing work, I am healthy.’

By suggesting extensions to Mrs Stone’s definition of ‘work’, he might be able to convince her that it took in her response to illness. In doing so, however, he would almost certainly need to discuss her husband’s dying and the alternatives to the ‘sitting’ to which he was condemned. Of course, it is also possible, in a negotiation of this kind, that Mrs Stone would convince the doctor that she was right, or that they would part in mutual respect for each other’s point of view. But to have achieved any of these results would be a success in communication.

(b) *Mrs Hannay*

Mrs Hannay was chosen because her thinking represents ‘illness as an occupation’. She did not immediately mention illness. In the early topics concerned with aspects of her residential history, Mrs Hannay expanded with relish on a number of perilous transactions involving her past houses or her furniture:

Q: ‘So you’ve really done quite a lot of entrepreneurial sorts of things off your own bat?’

A: ‘I could have done a hell of a lot more if I’d had some more freedom. Well actually my daughter, you see, always owned a part of the house, and she was more cautious than me, you see, and she always said “Oh Mother, I don’t think you should”.’

And later:

‘I’ve always been very *daring and adventurous*, and I’m very good in a crisis – I don’t panic very easily.’

As talk went on, though, it became plain that terms such as ‘daring’ and ‘adventurous’ were closely related to the words which she used to describe her experience of childhood illness:

Mrs Hannay: 'Well I learned to *give up* things, that was the first lesson, and I suppose I learned to *enjoy* things when they were available to be enjoyed. I've always *snatched at pleasure* when it was available . . . I've always *snatched the moment* and I've *never regretted it*, even if I've made horrible mistakes. I *don't regret* them, it's all part of living.'

Q: 'Yes . . .'

Mrs Hannay: 'And I've made some horrible mistakes I can tell you.'

Without expanding on the many different contexts in which these themes arose, it is perhaps possible to see that for Mrs Hannay:

One is adventurous if, and only if, one snatches at pleasure.

One gives up without regrets if, and only if, one snatches at pleasure.

If I am myself, then I am adventurous.

And she adds to these premises that:

If I am ill, I give up without regrets and snatch at pleasure.

Thus, Mrs Hannay gives her own particular charm and inflection to the crucial premises of 'illness as an occupation'.

If I am ill, then I fight, and fight well.

If I fight well, I cut losses and develop any resources I can find.

For without a doubt snatching the moment is Mrs Hannay's way of fighting the serious illnesses she has experienced.

'You see if you've had a really bad illness you realize what life is all about, the values of life. You've got to get your values sorted out, and how trivial all these (possessions) are. This is all just escape. It's all rubbish really. The real things in life are having good friends and health, and love if you can get it.'

But just as health in 'illness as an occupation' is not the opposite of illness, but a reserve from which the ill person gathers strength to fight, so for Mrs Hannay the health which is one of the real things in life is not the opposite of illness (from which she has never been wholly free), but a reserve from which one gathers determination. That reserve is only 'perfect' in the absence of illness, but it remains in the presence of illness. A comment, for instance, on an acquaintance:

Mrs Hannay: 'She's very determined; she's very strong; she's a woman who's had perfect health except for one illness.'

Q: 'Yes and quite a severe one too?'

A: 'Yes, it was, but then if she hadn't been well she'd never have come through it.'

At this point, then, I will set out the parallels so far established between Mrs Hannay's terms and those of 'illness as an occupation':

'Illness as an occupation'

being myself

being active

being healthy

being strong

cutting losses and developing resources

Mrs Hannay

what I've always been

being adventurous

being healthy/well

being strong/determined

giving up without regrets and snatching at pleasure

A first consequence of illness, then, is that Mrs Hannay remains what she has always been – adventurous.

As with Mrs Stone it is now necessary to consider Mrs Hannay's attitude to help, and for that purpose to ask whether she subscribes to a further premise, common to both systems of thought about illness which are considered here – the rule of reciprocity. Mrs Hannay's circumstances give a slightly different scope to the rule, but essentially she accepts it. For her its scope is in the area, vital to her, of maintaining friendships. She described at length, as a warning, how an old friend had become a complete bore.

Q: 'You've just described your friend – well now, what's your recipe for avoiding this?'

A: 'Well you ought to have some interest, you must have hobbies. Hobbies are absolutely essential, I think. If you don't have a hobby you might as well write yourself off'.

Similarly she described each of her friendships as being based on some interest, and implied that if one does not have interests, one has no friends. We may put it that:

Either one has an interest, or else one is written off.

as a parallel to

Either one repays help, or else one is a child.

but on the other hand .

If I snatch at pleasure, then I have an interest.

And so friendship is not impaired by illness, nor identity spoiled, for Mrs Hannay, although she recognizes the danger that a different response to illness could cause a sick person to be 'written off':

'And this seems to be her only interest: she dwells upon herself and her symptoms, you see, so that when she rings me up she just talks about her symptoms. Well you can imagine how boring that is for me.'

The consequences for friendship, help, and social identity of Mrs

Hannay's system of thought are paralleled by consequences for her relationship with doctors. 'Illness as an occupation' requires that:

If doctors explain, I find out what I'm fighting.
If I find out what I'm fighting, I fight well.

Mrs Hannay emphatically endorses these premises:

Q: 'What about the other difficult question, which is about whether people would like to know what's wrong with them?'

A: 'Oh yes.'

Q: 'Are you the sort of person . . . ?'

A: 'I would like to know, yes.'

Q: 'You would like to know?'

A: 'Oh definitely yes. I think it's insulting if you don't know.'

Q: 'And would you want to know not just what was wrong, what the diagnosis was, but also the prognosis?'

A: 'I'd love to know the prognosis, the life expectancy, the lot.'

Q: 'You would?'

A: 'Yes definitely, of course.'

Q: 'And why particularly?'

A: 'Well then I could cope with it. How can you cope with a thing if you don't know about it? You can't, can you? You can make your own decisions then, in the end.'

And the consequences, in terms of trouble for doctors, are equally not in doubt:

'I would do awfully in hospital now, and I really probably would be quite rude to them all . . . no I would be clever enough not to be *that*, that wouldn't pay. No, I would soft-soap all the nurses I expect, bribe them with stuff, and the doctors. Either just tell the truth to you . . . No, I think the attitude is all wrong towards the patients.'

Lest this be regarded entirely as imagination, it should be said that Mrs Hannay has been in hospital more than once and on one crucial occasion extracted a highly pessimistic diagnosis, prognosis and life expectancy from her doctor.

Conclusions on Mrs Hannay

The formal steps in the explanation which were treated at some length with Mrs Stone may be summarized more briefly with Mrs Hannay. The terms and implications of Mrs Hannay and their general equivalents are as follows:

'Illness as an occupation'

- 1 If I am myself I am active.
- 2 I have something to offer if, and only if, I am active.
- 3 If one does not have something to offer, one cannot repay help.
- 4 Either one repays help or else one is a child.
- 5 If I am healthy then I am strong.
- 6 If I am strong then I am active.
- 7 If I am ill, I fight and fight well.
- 8 If I fight well, I cut losses and develop any resources I can find.
- 9 If I fight, I am active.
- 10 If I do not demand explanations I do not find out what I'm fighting.
- 11 If I do not find out what I'm fighting I do not fight well.

Mrs Hannay

- 1 If I am myself I am adventurous.
- 2 If I am adventurous I have interests.
- 3 If one does not have interests, one has no friends.
- 4 Either one has friends, or one is written off.
- 5 If I am healthy then I am strong.
- 6 If I am strong then I am adventurous.
- If I am ill, I give up without regrets and snatch at pleasure.
- 9 If I snatch at pleasure, I am adventurous.
- 10 If I do not know about the diagnosis etc., how can I cope?
- 11 If I do not cope, I do not give up without regrets and snatch at pleasure.

I present this formal summary of Mrs Hannay's expressions as evidence that specific elements of her response to illness are explained by her system of thought, and that her system of thought is an instance of the general pattern 'illness as an occupation'. I argue that the explanation is valid because (a) the crucial premises (7/8) and (9) are not isolated in her thought but control a number of entrepreneurial activities which she has undertaken, and are therefore not rationalizations, and (b) the consequences entailed in the way of maintaining relationships with the healthy, seeking diagnostic information, and managing doctors, have been demonstrated.

The uses of eliciting this general pattern of explanation are less evident in Mrs Hannay's immediate history than they were in the case of Mrs Stone, for Mrs Hannay, as a 'normalizing' chronic patient, is what most GPs would admire and appreciate, an independent person willing to use her own judgment. She and her GP have come to a good understanding in which he explains his reasons for any change in regime which is needed, and she manages the regime efficiently by herself. Very possibly, though, if she were to be admitted to hospital with, say, suspected cancer, she would, on present evidence, come into collision with the routines for managing medical information in cancer wards. For if she kept to her principles, she would want to know the diagnosis,

prognosis, and life expectancy, and much of this would be liable to be kept from her. Concealment occurs because the pressures of routine on doctors, and their uncertainty about how to judge the 'personality' of their patients, prevent them from determining whether people really want to know even a bad prognosis when they ask for the facts. But at the same time doctors often speak as if they could tell whether people wanted to know the worst if only they knew the premises which had led them to ask.⁸ The general explanation invoked here seeks precisely to guide one to such premises. Mrs Hannay does not have suspected cancer, and with cancer in particular it is hazardous to predict reactions; but if she – or anybody else – were to say the kinds of things that have been quoted in the context of suspected cancer, it would be easy, on the present analysis, to locate her crucial premises, and it is plain that she would, in asking for the worst, be true to herself and to a life-long history of what she calls adventurousness.

Conclusion

I will end by drawing together the main features of this approach to the analysis of informants' accounts.

I began this paper with two conceptions of illness, as 'a destroyer' and as 'an occupation', which Claudine Herzlich distinguished in her interviews in Paris. I analysed these conceptions, and showed them to be logical systems of premises and consequences. The logic became apparent by the representation of each view of illness in sentences with a first person subject, such as might actually have been spoken by informants; and these autobiographical statements had the effect, not merely of representing informants' language, but also of making clear that the informant's language, and not the researcher's, was to be the means whereby the truth and falsehood of these particular propositions was decided. The researcher, I argued, would be successful when he had learnt to apply the informant's language and logic correctly.

To demonstrate the process of learning the informant's logic in constructing his conceptions about illness, and the means of moving on to an explanation of his words and actions, I presented two case studies from my own Scottish data of people whose thinking corresponded with Herzlich's two notions of illness. Learning the logic was revealed as a process of testing the contexts in which words and statements of the informant apply, of discovering terms which crossed several contexts, and of working out the relations of equivalence, contradiction and so forth between these terms. Once these aspects of the informant's language had been learnt, it became evident that both of these two

Scottish informants were putting forward a logical explanation of their thought and actions which, when economically arranged, bore the classic marks of a deductive theory.

The deductive rationale for his actions which an informant presents retrospectively was, I indicated, subject to certain tests. The rationale may be a rationalization if the premise is created to suit the consequence. Such premises are distinguished by the fact that they do no further work in the informant's life, either as the consequence of other premises or as the origin of other consequences. Therefore premises should be tested against the work they do in other aspects of the informant's life, and those that do no such work should not, for safety, be invoked in explanation.

A second test of the informant's deductive account lay in those consequences in thought and behaviour which confirmed it, but which were not the immediate subject of discussion when each premise was stated – whether because these consequences were in any case to occur on a future occasion, or because, although they had occurred, they were not relevant at first. Predictions through extending the field of relevance were tested with both informants, and one prediction about the future was tested with Mrs Stone. Both kinds of prediction are familiar from statements which we all make about people we know well, and they carry the proviso 'unless he changes his mind or holds a relevant qualifying premise which I do not know about'. The existence of this proviso does not detract from the value of the prediction or from the explanation which corresponds to it; rather it directs attention to possible reasons for failure should they occur.

It then became evident that the deductive accounts put forward by my two Scottish informants were essentially equivalent to the two deductive accounts of illness which I derived from Herzlich's Parisians. The test of equivalence was one of translatability and similar logical implication in the contexts discussed, and such a test should in principle be open to inspection. Once equivalences of this kind are established, explanations of thought and behaviour concerned with illness become possible not merely for the individual, but also for the general case. General explanations of this kind are appropriate to the practice of sociology, but it is not only general but also individual explanations which could be useful to patients and to practitioners in helping them to understand one another.

Much remains to be done to clarify the general implications of the method demonstrated, and in a second paper to follow I will select some of the more urgent practical issues for discussion. Some of these issues obviously concern research design. What topics, what methods of data collection, and what choice of informants, are appropriate

for logical analysis? Other issues concern problems of this type of explanation, and especially the problem posed by the person who holds two contradictory sets of premises – a situation which many of us suffer from much of the time. On this difficult subject of contradiction, I will resort once again to demonstration by case study.

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Notes

- 1 See e.g. W. J. Filstead (ed.), *Qualitative Methodology*, London: Markham, 1979; 'Proceedings of the SSRC symposium on the handling of qualitative data', *Sociological Review*, vol. 27, no. 4, 1979; G. W. Brown, Some thoughts on grounded theory, *Sociology*, vol. 7, pp. 1–16, 1973, for recent material.
- 2 Especially in America: see e.g. S. H. Matthews, *The Social World of Old Women: Management of Self-Identity*, London: Sage, 1979; A. Hochschild, *The Unexpected Community*, Englewood Cliffs: Prentice-Hall, 1973.
- 3 M. Johnson, 'That was your life: a biographical approach to later life', in J. M. A. Munnichs and W. J. A. van den Heuvel (eds), *Dependency or Interdependency in Old Age*, The Hague: Nijhoff, 1976.
- 4 C. Herzlich, *Health and Illness: A Social Psychological Analysis*, tr. H. D. Graham, London: Academic Press, 1973.
- 5 Readers unfamiliar with formal logic may be surprised at the relatively unqualified statement of each premise in what follows. This is because I have chosen for the sake of clarity to represent the sequence of thought vertically, so to speak, giving each element a line to itself; but it could equally be represented horizontally as a very long conditional sentence. Readers familiar with formal logic, particularly philosophers and ethnomethodologists, may wonder whether the method used here does or does not touch the fringe of certain long-disputed

problems, such as the falsifiability of propositions or the law of excluded middle. These points will receive discussion in a second paper.

- 6 The concepts of 'normalizing' and 'disassociation' occur under the same or different names in several authors, but I am thinking here of the relatively clear definition in F. Davis, *Passage through Crisis*, Indianapolis: Bobbs-Merrill, 1963.
- 7 Herzlich, *op. cit.*, p. 13.
- 8 J. McIntosh, *Communication and Awareness in a Cancer Ward*, London: Croom Helm, 1977, esp. ch. 2.

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R. G. A. Williams

Logical analysis as a qualitative method II: Conflict of ideas and the topic of illness

Abstract In the previous paper, a qualitative method was demonstrated which was based on logical analysis, and illustrated on themes arising in illness and old age. The emphasis was on revealing the deductions of informants, not on propounding a deductive theory about them. In the present paper general issues arising from the previous demonstration are considered. On issues of explanation and research design logical analysis is compared with statistical analysis. Then a crucial explanatory issue, concerning the person who combines conflicting ways of thinking, is worked through on a case study. Finally, the conclusions make further suggestions about the topics amenable to logical analysis, the contrast between logical and statistical explanation, the research methods appropriate to logical understanding, and the practical uses of the method.

Introduction

In the previous paper,¹ logical analysis was adapted for use as a qualitative method in sociology. As a qualitative method for analysing data, its aims are quite distinct from those usually propounded as an ideal for sociological theory; for, instead of helping to organize theory about objects, it can be used to display the order in reflections of subjects about themselves. By showing the implications of one conception for another in these reflections, logic can provide an explanation for beliefs and practice which is both powerful and concise. Furthermore the procedure involved in arriving at such explanations is clear and much of it is publicly accountable. Hence in both procedure and explanation, logical analysis may offer remedies for defects which have often been lamented in qualitative methods.

In the preceding paper, procedure and explanation, as they are conceived in this adaptation of logical analysis, were demonstrated by example, with only minimal discussion of general issues. The example taken was the way in which varying conceptions of illness in old age were explained by other conceptions which the old person entertained.

First, two coherent systems of thought about illness were analysed in the work of Claudine Herzlich — the systems which she labels 'illness as a destroyer' and 'illness as an occupation'. Then instances of these two systems were revealed in case studies of two Scottish widows from different social classes. Both seemed strikingly coherent and definite personalities, and this impression proved to derive from the logical coherence of their ways of thinking. Mrs Stone's thinking repeated the pattern 'illness as a destroyer', and Mrs Hannay's 'illness as an occupation'. Thus they provided a useful demonstration of the procedure and scope of logical explanation in the simplest possible cases.

In the present paper, the theme of old age and illness is again used as a demonstration, but the issues concerned are more general, and special attention is paid to the more complex case where one person entertains conflicting systems of thought. Intellectual conflict offers one of the most difficult and most interesting tests for the sensitivity of any method of explanation; and in order to understand the problem it poses for the present method, conflict must be related to the general conditions which are required if a logical explanation of informants' accounts is to be adequate. To appreciate, in turn, these general conditions of adequacy, it is helpful first to distinguish clearly between logical explanation, as it is used here, and the customary forms of explanation derived from statistical analysis. Logical and statistical analysis differ sharply at this level in their concepts of association of an appropriate universe, and of appropriate data and choice of informants. It is these differences which form the topic of the next section.

Logical and statistical explanation

Logical explanations, as they were used in my earlier paper, take the form of a set of premises held by one or more informants, including a definition of their practical situation, from which a set of consequences in their thought and behaviour follow, subject to conditions for (a) excluding rationalizations (b) supposing consistency with the operation of higher-order premises outside those considered, and (c) accepting the investigator's understanding and translations of informants' language. More will be said about the conditions mentioned later. As for what such sets of premises and consequences look like, the easiest thing is to refer to the figures at the end of the preceding paper. Ironically, this form of explanation may seem parallel with that prescribed for natural science, but instead of deriving from the thought of the scientist about predicted events, it derives from the thought of ordinary men

and women about intended ends, and the sociologist is essentially a midwife to that thought. Thus the analogy with natural science is, as usual in sociology, subtly misleading. Natural science operates logically on experimental or statistical associations of cause and effect. In the present type of sociological topic, however – though this is not true of all topics in sociology – logical connection can actually replace statistical association as the principle of analysis.

That the relation of premise and consequence to a subject is of a different order from the statistical association of objects perhaps needs no labouring. But there is a relation between truth tables in logic and contingency tables in statistics which illustrates the point neatly. While in logic a valid consequence *C* *always* follows from the same stated premises 'If *P* then *C*' and '*P*', the statistical relationship of these statements, as beliefs asserted by informants, may be highly ambiguous. In the following table, it is assumed that 'If *P* then *C*'; and '*P*' denotes the assertion of the relevant premises, 'not *P*' the denial of at least one of them, '*C*' the assertion of the valid consequence, and 'not *C*' its denial. If this table is read as a truth table, cells marked with an X represent true conjunctions of statements. If the table is then read as a contingency table, reporting the beliefs of informants, the X cells represent populations of self-consistent informants. In both the truth table and the contingency table the blank cell should contain no entry at all.

	P	not P
C	X	X
not C		X

If now, in the contingency table, some inconsistent informants do appear in the blank cell, as they usually do, and if in addition the top right-hand cell is populous, representing people who assert *C* for reasons other than those considered, which is perfectly acceptable, many surprisingly lukewarm levels of statistical association between *P* and *C* could occur. Indeed, much stranger paradoxes could occur in the nature of the statistical association than this, as I show in my conclusion. However, this variable association would occur without any ambiguity occurring in whether *C* was logically implied by *P*. That implication could be recognized even by inconsistent informants who assert both *P* and not *C*, and it plays a vital part, as I show later, in accounting for their behaviour. Thus an important explanatory relationship between *P* and *C* can be inaccessible to statistical analysis, in circumstances which are unlucky but not uncommon, while it remains accessible to logical analysis, and forms its chief subject matter.

Research designs based on statistical concepts are therefore

legitimately replaced, in dealing with the present range of topics, with designs based on logical concepts. A number of changes in procedure follow, as regards both data collection and choice of informants. The choice of informants in sociology is traditionally governed by the concept of a statistical population, but a statistical population is not the universe to which logical analysis aims to generalise – rather, it aims to generalize the relationships of logical elements. The idea of a logical element, and the forms of connection between logical elements, are quite distinct from the numbers and kinds of individual who use them as part of their thinking. Hence, according to whether the purpose is statistical or logical, either of two different kinds of generalization may be sought. Take the proposition that ‘I always wish the doctor to tell me the diagnosis’. For some purposes, as in considering policy about medical information, one may usefully ask *how many* people hold this proposition, and a statistical generalization will usually be involved in the answer. For other purposes, though, as in seeking reasons why people do not want to know the diagnosis, what is needed is a set of logical connections which relate to the proposition, and patterns of valid inferences will be the basis of the answer. Once these inferences have been drawn out and organized into logical systems there are again two kinds of questions, one logical, one statistical. If a statistical generalization is sought, it will estimate how many people entertain a system of thought, and what predictable association is found between that system and other attributes like age, sex, or class. On the other hand, if a logical generalization is sought, it will concern the equivalence or contradictoriness of systems of thought which are related, the ways in which contradiction between such systems are handled, and the capacity of each system to interpret new practical situations.

Thus what is done for natural causation by experiment or by establishing correlations within a population is done for logic by establishing premises and conclusions, and population characteristics of the persons who hold these premises and conclusions are not relevant to the validity of the systems of thought which emerge. It is only when further questions are asked, as for instance about how common or typical a particular system of thought is, that there is a need for informants to be statistically representative, to the extent required by the question which is being asked. Even then, the question may only require a broad answer, of a kind which shows merely that informants are not (1) systematically restricted in a way which is not required by the topic to be analysed or (2) unduly rare and atypical within the categories of that topic. The topic of Claudine Herzlich’s book, for instance, which was used as an extended illustration in the paper preceding this one, concerned

conceptions of illness in that part of the French population which is most active in forming ideologies. Herzlich's informants were statistically unrepresentative, being systematically restricted to about eighty members of the (predominantly Parisian middle class who could be interviewed conveniently. But the choice of the Parisian middle class was appropriate to the topic, and although the sample was one of convenience, it would be difficult to argue that their conceptions of illness were exotic or rare — and this is all that Herzlich's topic required.

The conclusions which have been drawn about the choice of informants also help to elucidate the kind of data required. I have suggested that not one but two kinds of explanatory generalization can be sought in social science. One kind involves statements that attributes are associated through some pattern of cause and effect; the other involves statements that a conclusion is consequent on a set of premises. I leave aside here the question of whether premises can properly be regarded as causes. Whether they can or not, a different procedure is proper for verifying the existence and the consequences of premises than that which is proper for verifying causes in the natural world. To grasp premises, a deep and evolving understanding must be established whereby sense is made by interviewer and informant together, and as this sense progresses so new and unforeseen questions can be asked on the basis of the sense already made. Moreover the validity of each sequence of reasoning produced by informants is tested first by the rules of meaning and logic understood in the use of their language (both interviewer and informant attempting to repair faults as they go along), and second by the place of that reasoning in the overall system that emerges (so that, as I will show later, rationalizations are definable in terms of the overall system — and so, of course, are contradictions). For this kind of data, personal involvement of the analyst in interviewing is indispensable, and the opportunity for repeated interviews and, ideally, observation of practice as a test of the meanings of words must be created.

Thus the research designs which are implied by logical analysis correspond broadly to those already favoured in qualitative methods. It is in analytic procedure that the principal novelties occur, and I will now turn to a closer examination of what is involved in meeting the three conditions which are necessary (though they may not in future be found sufficient) for an adequate explanation.

The conditions for a logical explanation

Adequate understanding and translation

As stated in the earlier paper, one condition for an adequate explanation

was that the investigator must have learnt the informant's language and logic; and a second related condition was that the investigator's translations of informants' terms into general terms must be acceptable.

The translations made between informants' terms and general terms are less problematic, in the sense that with small samples they are more easily made open to inspection, and each reader can then use his own knowledge of the language to appraise the result. Less accessible is the process of learning the informant's logic, and I concentrate here on questions in this area.

In saying that he has learnt the informant's logic, the investigator makes claims not primarily about the dictionary meanings of words, but rather about the contexts in which the informant's terms are used. It is not safe to assume that an informant is speaking within public conventions known in advance to the researcher, and the case of Mrs Stone, described in my earlier paper, is a warning against this assumption.² If dictionary definitions are applied to what Mrs Stone said about help after bereavement, it is hard to escape a contradiction, but if her personal use of language in such contexts is invoked the difficulty disappears. A striking feature of her language is its vagueness in literary terms — a great deal of work is done by phrases like 'sit down' or 'go out' which require interpretation by the context. Consequently, it is in Mrs Stone's own terms and contexts that the systematic implications of her thinking must be demonstrated, and it is in her terms and contexts that truth or falsity must be assigned to her propositions.

First of all, then, the investigator claims that the informant's terms and propositions as expressed in their conversation will be relevant again at other times and in other settings understood by the investigator. The claim thus made can be supported by the evidence of return visits, of what the informant is reported to say to other people, and of observation, although much of this would be tedious to report. Instead, a principle of selective reporting can often be found, and I make a suggestion about this at the end of this section.

However, as well as learning what terms and propositions apply to what contexts, the investigator also claims to have learnt what consequences follow for the informant in that context. The claim is easily summarized by presenting the informant's natural language and rearranging it in logical order. However, such rearrangements in formal terms have certain traps. The implications of a formal proposition are precise and may be more (or less) extensive, depending on the formal system involved, than the plain English meaning of its apparent equivalent as it emerges from a particular context.³ If the formal implications are fewer than the context suggests, the problem is only that

the analytic reconstruction will be less determinate than it need. But that is usually better than being more determinate than it should. Hence the crucial danger lies in the hidden extra implications of a formal system.

One of the implications of most formal systems about which there has been some discussion is the law of excluded middle, and objections have been raised (often by ethnomethodologists) against accepting the implications of that theorem in ordinary speech,⁴ on the grounds that the meaning of words is always ultimately indeterminate. This indeterminacy is not news to logicians, some of whom use it as ground for urging that the law of excluded middle is still more necessary to observe as far as possible.⁵ But while a counsel of that kind may fairly be urged on academics, it is perhaps unwise to assume that it is always the practice of the laity to speak in such a way that all the provable theorems of formal logic are understood and intended. The law of excluded middle, for instance, is derived by invoking two rules which ordinary speakers may seldom use: the rule of double negation, which states that to deny the falsehood of P is to assert P, and the rule of *reductio ad absurdum*, which states that if a contradiction is derived from an assumption, that assumption is, on any remaining assumptions, false. Unless inferences of this kind are explicitly drawn by informants, it may be unwise to attribute them. However, problems such as these do not mean that any possibility of attributing formal implications is thereby excluded. Some formal implications must be attributed in order to understand language at all.⁶ Implications of this kind may therefore be freely invoked, and only if more complex entailments seem to be suggested should one check that they are accepted. Since, however, the rules of implication followed in presenting a formal rearrangement of informant's language are apparent in the presentation, no great problem of accountability arises here.

The main problem, therefore, in showing that adequate understanding has been achieved, lies not so much in the implications as in the contextual relevance of the informant's propositions. The investigator, of course, may have checked his understanding of contextual relevance to his own satisfaction through the means described earlier (revisits, observation, corroborating reports, etc.), but the problem lies in when he should show this checking process for inspection. Because checks of this kind are a constant preoccupation in qualitative field work, and the details to be checked are so diffuse, some principle of selection is necessary. Perhaps the critical cases where checks should be shown are those where the investigator wishes to assert that the informant has shifted the application of a premise or consequence – whether to avoid a contradiction or to help on a conclusion. In cases of this sort, if the

informant disavows the problem, the investigator's credibility lies partly in the previous checks on the application of terms and propositions which he has made.

This crucial test of adequate understanding may sound as if it runs against the principle of learning the informant's logic, for such sustained disagreement with the informant runs against the grain of what, up to this point, is usually thought of as good field work. In a disagreement thus far, the informant is always right. But after a certain point, an investigator who thinks that an informant's premises explain and predict his judgments and actions must check for any shifts in the application of those premises or consequences which permit their consistency to seem greater than it really is. This requirement arises however, from a condition for the validity of a logical explanation which is dealt with in the next section.

Eliminating rationalizations

So many inappropriate modes of rationality have been foisted on the public, and the discrepancies of public behaviour from these models are so apparent, that it is common in sociology to suspect that most logical accounts of action are 'rationalizations'. A useful corrective, however, is Schutz's notion of common-sense rationality,⁷ and the aim here is to distinguish this logic which does commonsense work from the logic which is merely decorative.

All logic can be thought merely decorative in that there is nothing about his initial premises which *necessitates* that a person acts in a particular way; for if he wants to, he can simply change his premises. When the change can be made at will it is what we call 'rationalization'. But some changes of premises bear a cost, in making other favoured conclusions impossible. If this cost is accepted and avowed, then the change which gives rise to it is not a rationalization, in the sense which I will be using it, and it is premises of this pivotal variety which are particularly interesting.

These pivotal premises explain and predict their consequences so long as they remain unchanged. However, these premises do not become useless in explanation even if they are changed, so long as this change is an avowed response to experience. Rather, they explain the new consequences also when taken together with the new premises and the event which initiates the change. Moreover, in so far as the initiating event and its effect on the old premises are predictable, the old premises and the event leading to the new premises not only explain the new consequences but also predict them. To illustrate with a very obvious example: if I hold that I have an illness which is incurable, and that I should consult doctors for what is curable, but not for what is

incurable, and if it can be predicted that the cure of a fellow-sufferer known to me will cause me to change my opinion about my illness, then it can be predicted that, given my premises and the cure, I will consult a doctor. This mixed prediction, dependent on events affecting premises, and the purely logical one by which one can predict that, my premises remaining constant, I will not consult a doctor, form two fundamental patterns of valid explanation and prediction, and it is against these patterns that the interference of rationalization may be recognized.

By thus identifying what is not a rationalization, two ways in which rationalization occurs now become evident. The first way is when a premise is invented *ad hoc* to suit a specific consequence already actual or desired in itself. Such inventions can be ruled out by testing the premise against other premises or consequences which were not immediately relevant when the premise was stated, or which arise subsequently. The testable premises are therefore usually those which have antecedents or consequences in several diverse contexts.

The second way in which rationalization occurs is the one referred to in the previous section, where consistency is maintained by an unavowed shift in the application of premise or consequence. This kind of unavowed shift is tested either by contrasting the new context to which the shifting term or proposition is applied with the previous ones (as was described in the last section), or else it is tested as a proposition by the propositions which earlier entailed it or by those which it earlier entailed. This latter test by entailments is again made possible by premises which have several antecedents or consequences.

An example may make clear how both the possibilities of rationalisation are tested in explaining responses to illness. In the conceptions of illness described by Herzlich and analysed in my earlier paper, a crucial differentiation of premises lay in whether or not it was accepted that

If I am active, then I am not ill.

If this proposition is not employed by the same person in other contexts, then it may be a rationalization in the sense of being a casual invention. That it was not a casual invention for Herzlich's informants, however, is clear from the way in which 'activity' was defined in other contexts. For some people, above all, activity was *work*. 'I've worked all my life since I was young, and now . . . ' said one of Herzlich's respondents for whom illness is 'a destroyer'. In other words, for a person of this way of thinking there exists a subsystem of ideas, separate from the subsystem concerned with illness, for which the proposition 'If I am active, then I am not ill' takes on a particular value. A very obvious version of such a subsystem might run as follows:

I am active if, and only if, I am working
 If I am working, then I am happy
 If I am working, then I am out on the shop floor

on the other hand

If I am ill, then I am not out on the shop floor

hence

If I am active, then I am not ill

Here the premise of 'illness as a destroyer' is in fact the consequence of another valued subsystem of thought, and is not therefore a casual invention.

Taking the same premise of 'illness as a destroyer', one can then imagine a subsequent situation in which an informant maintained the consistency of this premise and its consequence by an unavowed shift in the application of one of them. This situation might be one in which the informant declares that he is ill and therefore inactive while at the same time being, as far as the investigator can see, active. One way in which the investigator can be sure that a shift in the application of the word 'active' has in fact occurred is if the informant accepts that he is still working, for if he does then this implies that he is active in the sense he originally used, but has shifted the application in order to be consistent with his sense of being ill. This test by implication is thus a second benefit derived from attending to pivotal premises. This second test, and the earlier test which establishes that premises are pivotal and not casual inventions, together provide a means for excluding rationalizations and establishing the validity of a logical explanation.

Testing and accounting for contradictions

When dealing in my earlier paper with people who thought and acted consistently, I stated the third condition for explaining their responses as merely supposing consistency with the operation of higher-order premises than those listed in the system. But there is more to it than this. A higher-order premise not elicited by questioning will only manifest itself by compelling, in certain instances, a decision contradictory to the consequences implied by the premises which are elicited by questioning. If no such contradiction arises, and behaviour and belief continue in accordance with the system, consistency with the operation of higher-order premises is assumed until further notice, and no more need be said than that the discovery of an informant's higher-order premises may one day come in useful, and that those who attempt a deeper inquiry of this kind will be wiser in the long run. In the short term, the explanation offered by even an obviously limited

subsystem of thought will be adequate to the purpose in hand. The problem and opportunity for further explanation, then, arises essentially when behaviour or belief is in contradiction with the consequences of the system of premises listed.

A more general statement of the third condition, therefore, requires the explanation of contradictions, and contradictions arise from many sources. Trivial contradictions may arise from misunderstanding informant's language, as was shown in the previous paper in the case of Mrs Stone.⁸ Not so trivial, but not problematic, is the case just mentioned where a higher-order premise demands conformity to a subsystem of thought which has not been considered, resulting in the system under consideration being overridden. If both these initial possibilities are ruled out, we have a situation described by Bernard Williams as a 'confrontation' between two systems of thought; and if what I have called rationalizations have been ruled out, too, we have something close to what Bernard Williams calls a 'real confrontation'.⁹ As he remarks, though, a confrontation will not necessarily be perceived by the person who experiences it, or alternatively it may exist only in his imagination through a misunderstanding. Both these cases, however, together with the case where confrontation is fully 'real' in Bernard Williams's sense, present an important problem of logical explanation; for the first task of logical analysis in such cases, in addition to revealing the two systems of thought which are in contradiction, is to show the conceptual responses which people make to the contradiction, and in particular to elicit whether these responses are themselves based on premises relevant to resolving contradictions, or are the outcome of situational influences.

To show how 'confrontations', as the most serious and interesting type of contradiction, may be analysed and explained, I return again to the method of demonstration.

The logical analysis of mental conflict – a case study

I have indicated that analytically the difficult but interesting case is somebody who entertains two systems of thought at the same time, although they are at some point contradictory. In the long run there can be few people who do not experience such a situation, since any unexpected event may challenge one's thinking, and the resultant contradictions may be an essential element in stimulating a more developed system of thought which can cope with the new experience. If, therefore, the present method of analysis is able to deal with the contradiction of systems, it can also contribute to an understanding,

through the longer-run aspects of a career study, of the genesis of these contradictions and their resolution. To illustrate the method in the simplest case, I have chosen an instance of real and acknowledged confrontation between two systems of thought in the same person. That this kind of contradiction can be handled, and its genesis and resolution illuminated, is suggested by the case of Mr Grant.

Mr Grant has recently retired after working very intensely up to the moment of his retirement; he had little chance to prepare for the change, and the experience has left him without a sense, in his own word, of 'pattern'. He might just have meant by 'pattern' something like routine, but it was clear from other contexts that it included something more profound. A considerable reader, he attempted constantly to make an explicit logical pattern out of problems arising in different contexts of his life, using literary references to aid him. Anybody literary or academic is especially prone to rationalize, in the sense of developing sets of premises which do little work in their own lives, and Mr Grant was no exception. But during this period he was also struggling to resolve puzzles in these patterns which arose directly from his experience. His patterns and his puzzles were wide-ranging, and here I focus only on the elements which are related to the systems of thought about illness which are under discussion.

Mr Grant did not wish to be tape-recorded, although he knew I would make notes, and what follows is therefore seldom given quotation marks unless I was convinced in making notes that I remembered an exact sentence.¹⁰

In his general view of what it was to be ill or disabled, Mr Grant was clearly on the side of 'illness as an occupation'. He instanced thalidomide children as cases of how you can turn apparent disadvantage to advantage: if you're blind you develop better hearing or smell, if you've a withered hand you learn to do more with the other. By contrast he described an elderly friend who was competing with her husband to see which could be the real invalid. He went to see her in hospital and said 'Look Rachel, if you don't pull yourself together you'll be here for good. It's up to you, the doctors and nurses can't do it for you. Whether or not his intervention made the difference, she did pull herself together, and recovered rapidly. Clearly in all this he considers that:

If I am ill, I fight and fight well.

If I fight well, I cut losses and develop any resources I can find.

At the same time he spoke of people being healthier as the result of a balanced life of mixed activity, and of health as something on which strains could be put by excessive desk work — strains evidently not

equivalent to illness, but rendering one liable, if one left oneself open too long, to invasion by illness. In all this there is a clear indication of health as a reserve, and of the propositions:

If I am healthy, I am strong.

If I am strong, I am active.

or to be more precise of the stronger proposition:

I am healthy if, and only if, I am strong and active.

As for the other two distinctive premises of 'illness as an occupation', Mr Grant approved, and hoped that he would exemplify, the propositions that:

If doctors explain, I find out what I'm fighting.

If I find out what I'm fighting, I fight well.

He was realistic, and did not know whether he would match up to these ideas, but there was some advantage, Mr Grant felt, in knowing a fatal prognosis, and he did not think an intelligent person would *never* know. This advantage he exemplified by the hero of Neville Shute's novel *The Chequerboard*, who knew he had a year to live and planned it all quite differently from what he would have done if he hadn't known. A close relative of Mr Grant's does know her prognosis (a pessimistic one in the long term) and she thinks that to know has advantages.

In all respects, then, Mr Grant advanced premises which exemplify 'illness as an occupation' in its distinctive features. It was the general system of thought which he approved, and he had tested it against the experience of his friends and relatives and against literary examples.

However, retirement had presented Mr Grant with an experience for which he could not account to his own satisfaction. He found himself, in his own words, 'easily fatigued' and 'not content', and it was clear from the responses he had made to this experience over a few years that these mild terms covered something sufficient to be disturbing. He was not accustomed to thinking of himself in this way. The anomaly was moreover constantly brought home to him by people stopping him in the street to ask, 'How are you enjoying retirement?' His inference here was elliptical and striking:

'I wasn't *a person in my own right*, I was the retired solicitor'.

So painful was this inference that he ascribed to it his decision to move elsewhere. What was the reasoning, though, behind his inference? A possible line of connection would run as follows:

I am the retired solicitor.

How am I enjoying retirement?

I can reply to that question in public only if I enjoy myself.

I enjoy myself if, and only if, I am active.

If I am myself I am active.

But I am not active, I am fatigued.

I am not myself.

I am the retired solicitor.

Whatever the line of connection, though, Mr Grant's experience had produced a consequence which, in the discussion so far, has only been produced by 'illness as a destroyer', namely

I am not myself.

Did he also subscribe to the premise distinctive of 'illness as a destroyer' namely

If I am active, I am not ill?

The answer to this question requires some sifting.

It is relevant to the notion of 'illness as a destroyer' that when I first met him, Mr Grant was painfully considering some sequences of thought which later he was to reject as a 'slippery slope'. It came to him that being regarded as retired, he was perhaps regarded as old, and he identified his difficulty in retirement with the words 'I don't *feel* old, that's the trouble', and added that he couldn't get used to the idea that 'the body' could not do as much as it used to. One way of putting this might be:

Is it the case that, if I am not active, I am old?

Alternatively, to capture something of Mr Grant's aloof reference to 'the body' declining in activity one might put it something like this:

Is it the case that, if the body is breaking down, I am old?

However the suggestion thus derived of investigating his physical condition would probably not have been decisive, had he not also held a number of premises on these topics which, when combined with these, led inevitably to the conclusion:

The body is breaking down.

Mr Grant maintained forcefully how mistaken his doctor and his friends had been in their plans for old age, plans which in both cases involved great activity in setting up a large garden in the country. He reasoned against them — and against his own love of gardening — as follows:

'Three score years and ten — and that's curtains. That leaves me perhaps four or five years of active life. It doesn't make sense.'

Now Mr Grant is sixty-three. Subtracting rapidly, one sees that two to three years are unaccounted for, or more precisely, they are accounted tacitly to *inactive* life. Taxed with this interpretation, Mr Grant confirmed it. Indeed his friend's experience gave rise to the interpretation, for a stroke had incapacitated the friend at the height of his gardening activity. The rapidity with which Mr Grant had generalized this instance, on which he dwelt at some length, reveals in the back of his mind the premises:

If I am old, I am not active.

If I am old, the body breaks down.

The consequence of these and the way he put the relation of age and activity earlier is two two-way propositions:

I am old if, and only if, I am not active.

I am old if, and only if, the body breaks down.

Amongst the consequences of these ideas in turn are, first, something very close to the by now familiar premise distinctive of 'illness as a destroyer':

If I am active, the body is not breaking down.

and also a new corollary

If I am not active, the body *is* breaking down.

The consequence of this last corollary, given his experience of being fatigued since his retirement, is clearly:

The body is breaking down.

This has been a complex analysis for the obvious reason that Mr Grant was in a complicated state of mind, but its outlines may now be summarized quite easily. Simply put, Mr Grant's conception of illness and disability by themselves matched 'illness as an occupation', while his conception of old age matched 'illness as a destroyer'. Retirement had precipitated an intense conflict in him over whether he was now old or not. Central to this conflict was the meaning of his sense of fatigue, of discontent, and of not being himself. It appeared from his conception of old age that a sense of fatigue and inactivity must mean that he *was* old. But in that case the body was breaking down, and he was also therefore ill.

At this point there opened up for him the escape route provided by the premises of 'illness as an occupation'. If he was ill, he would fight, and in order to fight he would find out the diagnosis. In this way he could be himself again.

Mr Grant used his escape route with all the thoroughness that a professional mind could bring to it. He decided he would secure specialist advice, and waited until after he had moved house into the area of a teaching hospital before going to his GP. Apart from fatigue he could produce a specific symptom – pain in a particular joint – but whether or not this helped to persuade the GP, the pain, from the way he mentioned it, was clearly of no consequence to Mr Grant. But he got his referral to the specialist, and a number of X-rays to boot. At the end of all this, all tests proved negative.

For Mr Grant, this outcome was an unqualified success. He referred to the whole process, significantly, as an ‘MOT’ test, and rightly so, for its purpose was precisely to certify him as road-worthy for a period of time to come. The body was not breaking down, therefore he was not old. Consequently he could now reject the premise which had got him into this situation:

If I am not active, I am old.

As he put it, he had been prevented from going down ‘the slippery slope of “oh-I’m-getting-older-I’m-not-so-active-as-I-used-to-be”’. The premises of ‘illness as an occupation’ had proved triumphant; the premises of ‘illness as a destroyer’ were not relevant to him but to others. He quoted Bob Hope:

‘Always accentuate the positive feeling –
Don’t mess with Mr. In Between.’

Conclusions on Mr Grant

It is beside the point with Mr Grant to set out a formal recapitulation of the general patterns of which his thinking provides instances, since the aim here has been to show how he entertains two partly conflicting systems. One would not expect in this case to demonstrate consistent consequences. Instead, the focus shifts to identifying those premises of either system which are most liable to be overturned in typical situations of conflict, and to showing what situational and logical factors affect the outcome of the struggle between these premises. The painfulness of the struggle in Mr Grant’s case is sufficient evidence that the conflict was not between mere rationalizations; and if it were insufficient evidence, the test which has been suggested previously could be adduced, that is, the part played in other contexts of his life by the distinctive premises which controlled his behaviour in illness. Here though, I will simply summarize the premises in conflict.

Mr Grant’s conception of old age and his conception of illness collided when his conception of old age suggested:

If I am not active, I am old.

while his conception of illness suggested merely

If I am not active, I am not healthy.

Not that these propositions are contradictory in themselves, but, as was demonstrated earlier, the systems to which they belong lead to contradictory consequences such as refusal of, as opposed to eagerness for, a medical diagnosis. The collision of systems came at the premises identified because Mr Grant wanted to know what to make of an ambiguous phenomenon – his fatigue. Acting like the ideal natural scientist, he deduced from his conception of old age the testable conclusion:

The body is breaking down.

But he would not have tested this conclusion if he had continued thinking within the sequences of 'illness as a destroyer'. It is here that his ability to shift back to 'illness as an occupation' supplied him with an escape route, enabling him to test and reject the relevance of his conception of old age by obtaining a thorough specialist examination. Thus he could conclude:

If I am not active, I am not healthy.

And within his concept of health as a reserve of strength this meant simply:

I've emptied the bucket and it's got to fill again.

In this explanation of how Mr Grant resolved his conflict no situational factors need to be presupposed, although the evolution of further questions might lead to their introduction. For the factors which enabled him to solve the conflict were themselves premises relevant to deciding contradictions. These premises concerned the necessity in this situation of testing a decisive hypothesis – not 'messing with Mr. In Between' – and of the superior thoroughness of tests performed by specialists.

The possible uses of analysing Mr Grant's way of thinking are various and intriguing. How far were his doctors aware of the nature of the experiment they were engaged in? How far, if they were aware of it, would they approve of it as appropriate medical work? His general practitioner, at any rate, seems to have grasped something of his problem, for Mr Grant attributed to him a statement that the last five years had taken it out of his body, and he must wait for it to recover; in other words the GP sought to confirm the premise

If I am not active, I am not healthy.

or — more precisely

If I am not active, my reserve of health is low.

On the other hand the GP does not appear to have taken in Mr Grant's conception of old age. He talked to Mr Grant about his own retirement plans for a large garden, perhaps assuming that in this way he could suggest new interests to build up his patient's reserves. Mr Grant's reaction was simply:

'Fancy him making that mistake.'

It is likely, for all the present reprieve, that Mr Grant's conception of old age will one day be a problem to him. But the reader may agree that, while his GP could usefully have tackled this conception, his chances of grasping its existence without knowing some general patterns of reasoning which might produce Mr Grant's enigmatic response were slight. Moreover, in what he did do, the GP was dealing unusually successfully and sensitively with what is usually termed a 'social' problem. In other areas of medicine, Mr Grant stood to be dismissed as neurotic and hypochondriacal. I hope at the minimum to have shown that he did not deserve these labels, and that his GP's way of handling his case was the only one likely to clear up the problem.

Summary and conclusion

Logical analysis has been put forward as a qualitative method suitable for certain topics, in particular for revealing the interrelation of definitions, beliefs or evaluations, whether individual or cultural; and its uses on such topics have been illustrated by material from informants in France and Scotland. I suggested that Claudine Herzlich's analysis of the French material was capable of generalization by logical methods, and in that form it offered two accurate models of the way in which Scottish informants reasoned about illness, their present values, and their conclusions from biographical experience. Those two models do not, of course, exhaust ways of thinking about illness. Nor do they exclude the possibility that individuals, by introducing an unexpected variant for a standard premise, may create somewhat idiosyncratic versions of each model. Much remains to be explored. But basic patterns of reasoning on a topic — the standard arguments — do not perhaps vary so much at one time and place that they cannot be understood in terms of certain dominant systems. Greater variation probably occurs in the contradictions between systems which individuals entertain; but here too a knowledge of the systems in conflict may reveal a

degree of order in the way specific contradictions occur. In this paper only one kind of contradiction has been explored, a contradiction both real, and recognized by the informant. The exploration of unrecognized contradictions is perhaps the next task, and several other forms of contradiction have been mentioned whose analysis will need to be worked through in time. In developing each of these topics, the aim of logical analysis as it has been used here will be to reveal the order in the thinking of ordinary people about their own practical situation, and in the interrelation of the ideas which inform their decisions about themselves.

Although the interrelation of ideas is the most obvious subject for logical interpretation, I should add that the analysis is capable of more than has been suggested here. While meanings, as opposed to situational factors, have been the present focus, a definition of the situation, such as 'I am ill', has in each case study been the intermediary between a person's general premises and the consequences which have sprung from them. Changes in the definition of the situation incur new consequences, and may result either in the acceptance of those consequences or the rejection of premises. A consideration of these outcomes provides a clear analytical basis for 'career' studies, and where premises are rejected there is also an opportunity to bring in concepts from the theory of cognitive dissonance.¹¹ Another opportunity provided by considering people's definitions of their situation is analysis of the relation, logical or statistical, between their particular situations and their adherence to a system of thought. The relations between aspects of class and culture, or between religious or philosophical ideas and elements of the social order, are subjects to which logical analysis, in conjunction with the usual analyses of social attributes, might contribute.

The implications of the method for research design were discussed briefly in the second section on 'logical and statistical explanation'. The distinction there made between logical entailment and statistical association can now be taken a little further in the light of the three case studies presented in this paper and its predecessor. I showed earlier that measures of statistical association would often give only a poor reflection of the association which occurs through a typical mode of logical implication. The origin of the difficulty lies in the contradictory position of such people as Mr Grant. The preceding paper dealt only with coherent cases, one of whom, Mrs Stone, represented the entailments of 'illness as a destroyer', while the other, Mrs Hannay, represented 'illness as an occupation'. Suppose now that in addition to the Mrs Stones and Mrs Hannays, and others, who, as was indicated earlier, may act like them for different reasons, there is a significant population of Mr Grants — as very likely there is. To represent these populations I

return to the contingency table exemplified in the second section, and this time 'P' will represent the premises of 'illness as a destroyer', 'not P' the absence of at least one of them, 'C' the logical consequences of 'P', and not C the absence of some of those consequences, thus:

	P	not P
C		
not C		

All the Mrs Stones would go in the top left-hand cell, and all the Mrs Hannays in the bottom right-hand cell, producing, so far, a perfect statistical relationship. However, those who act like Mrs Stone for different reasons would go into the top right-hand cell; and more seriously, all the Mr Grants would have to go in the bottom left-hand cell. With enough Mr Grants, and enough people who act like Mrs Stone for different reasons, one could banish all apparent relation between premises and consequence, or even produce a negative relation. Yet, despite the statistical appearances, the relation of Mr Grant's premises to his consequences is as strict and inevitable as that of everyone else in the table. From a logical point of view, moreover, Mr Grant is perhaps the most interesting of the three cases, and the one for whom logical analysis has the most important function.

The sharp difference which is thus apparent between logical and statistical analysis reinforces the point that in the logical mode a less stringent choice of informants is required, but a more complex and intensive form of interviewing. On this interviewing depend conditions for the adequacy of explanation and prediction which have been discussed at some length, and which are comparable with the conditions which are recognized in statistical analysis. Such interviewing should ideally include some participant observation, but in any event should involve acquiring skill in, and respect for, the informant's own language. I noted, for instance, in my first case study, that a feature of Mrs Stone's language was its vagueness and its reliance on an assumed context. A great deal of meaning was borne by expressions like 'sitting down' and 'going out' which would be intolerably loose in a questionnaire. Further, if dictionary definitions are imposed on someone like Mrs Stone, they are capable, as was shown earlier, of generating contradictions which are hard for her, and also for her interlocutor, to solve. It would be easy from this to conclude that Mrs Stone didn't know a contradiction when she saw one, and that a real explanation of her behaviour required the researcher to define it for her. I hope I have shown that this conclusion would be seriously mistaken, and would be

the product of a method unsuited to eliciting premises, however suited it is, and has been, to an analysis of situational causes.

To conclude this review of the implications of logical analysis, some general points may be made about its possible practical uses. Each case study ended with a brief account of the advantage which might have been gained if doctors had understood the system of thought which each informant was using. But how would they, or any other professionals, acquire that understanding? First, they might gain something from knowing the nature of common systems of thinking about illness. That knowledge would lead them to suspect the existence of premises in a patient's thinking which were not the immediate focus of conversation. Second, to make proper use of that knowledge, it would be necessary to acquire the skill of eliciting premises as well. Some do this naturally, but others, even if they do not grudge the time, do not always know how to make use of the conversations they have. However, it is possible to acquire such skills, and the advantage, both for laymen and for professionals, is obvious.

Taking chronic illness in old age in particular, logical understanding permits co-operation rather than the use of authority, and in chronic illness co-operation is usually a great deal more effective. Thus to understand something of the situational determinants of an ill old person's actions is not enough, for this leads only to attempts to manage or control him. Even when such attempts are successful, they may inflict an inner injury in dividing behaviour from the mental premises which are the source of self-determination. By contrast, perhaps the most potent influence on co-operation is the recognition of the other person's premises, and the demonstration that a course of behaviour flows from them.

It may be objected that in most of the organizations which deal with old people, staff are often too rushed for more than brief exchanges. This is true, but provided that at some point in the past an understanding has been gained of an old person's system of thought, recognition of the premise which is guiding the behaviour of the moment can be something instantaneous. A brilliant portrayal of such an instant, drawn by a playwright who was himself a doctor, occurs in Chekhov's play *Uncle Vanya*, when the ill old Professor Serebriakov won't go to bed.¹² He has been thoroughly difficult to his second wife and his daughter Sonya for much of the night, and has worn out all his helpers. At this juncture the old family nurse enters. Essentially, she recognizes that behind his behaviour lies a definition of his illness established in relation to his dead wife Vera Petrovna. In comparison with what has gone before, the exchange on stage is extraordinarily brief and dramatic, beginning with Serebriakov, who is quite furious.

Serebriakov: Everyone's awake, everyone's worn out — only I am thoroughly enjoying myself.

Marina (going up to Serebriakov tenderly): What is it my dear? Your pain again? I've got a grumbling pain in my legs, too, such a pain! (tucks the rug in) it's that old trouble of yours . . . Vera Petrovna, Sonya's mother, used to get so upset about it — she couldn't sleep of nights . . . ever so fond of you she was . . . (a pause) . . . The old are just the same as the little ones, they like someone to pity them — but nobody pities the old. (Kisses Serebriakov on the shoulder). Come to bed, my dear . . . come, darling . . . I'll give you some lime-flower tea and warm your feet . . . and say a prayer for you . . .

Serebriakov (moved): Let us go, Marina.

Marina: I've got such a grumbling pain in my legs myself, such a grumbler! (She and Sonya lead him off). Vera Petrovna used to get so upset she used to cry over you . . . You were little, then, Sonya, you didn't understand . . . Come along, come along, sir.

(Serebriakov, Sonya and Marina go out)

If one may precis this passage (and it is already so succinct that a precis is a violation), it is as if the nurse — and because she 'understands', unlike Sonya, she can do this — recognizes in Serebriakov's sentence the following premises and conclusion:

This pain is the one which is pitied by my wife.

This pain is not pitied by anybody living.

My wife is dead.

And having shown that she recognizes the source of his upset by expressing it, she starts from a similar premise to a different, and calming, conclusion:

This pain is the one which was pitied by your wife.

Pain 'grumbles' and deserves pity at every age.

Your wife was right to pity it.

The whole transaction lasts about a minute. The average general practitioner, if he or she could perform such a miracle, would have five minutes to spare for the next patient.

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Notes

- 1 See the previous paper in this issue.
- 2 Ibid., p. 152.
- 3 Standard warnings on the relation of natural sentences to formal logic can be found in most textbooks. G. J. Massey, *Understanding Symbolic Logic*, New York: Harper & Row, 1970, devotes a chapter (ch. 20) to the subject. A standard text on the mechanics of formal logic is E. J. Lemmon, *Beginning Logic*, London: Nelson, 1965.
- 4 H. Mehan and H. Wood, *The Reality of Ethnomethodology*, London: Wiley, 1975, esp. pp. 64 f.
- 5 A. Kaplan, *The Conduct of Inquiry*, San Francisco, Chandler, 1964, esp. pp. 65 f., is sympathetic to some of the lines later taken by ethnomethodologists, while reaffirming the law of the excluded middle.
- 6 A number of confusions in this difficult area are disposed of in S. Lukes, 'Some problems about rationality', *Archives Européennes de Sociologie*, vol. 8, pp. 247–64, and M. Hollis, *Models of Man*, Cambridge University Press, 1977, pp. 145 f., who regard the law of non-contradiction and some others, but not the law of excluded middle, as a basic requirement of translation.
- 7 A. Schutz, 'Common-sense and scientific interpretation of human action', in *Collected Papers*, vol. I, ed. M. Natanson, The Hague: Nijhoff, 1971.
- 8 See above, note 1, p. 152.
- 9 B. A. O. Williams, 'The truth in relativism', *Proceedings of the Aristotelian Society*, vol. 75, pp. 215–28, especially pp. 217, 221, 223.
- 10 Generally, I am guided in my attributions by an experiment in which I compared field notes and tape recordings, both made on the same interview, in a small set of test cases.
- 11 L. Festinger, *A Theory of Cognitive Dissonance*, London: Tavistock, 1957.
- 12 A. Chekhov, *Plays*, tr. Elisaveta Fen, London, Penguin, 1954.

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